

# STAR: A Multi-Purpose Wearable Gadget from India



Anagha P. is a technical correspondent at EFY

**W**earable technology has seen tremendous growth in the past year. A recent report on wearable electronics market by *Business Insider* forecasts that the global wearable devices market will grow at a compound annual rate of 35 per cent over the next five years, reaching 148 million units shipped annually in 2019. Competing with global players in this arena, we now have a wearable band designed and manufactured completely in India.

Founded in 2013, Belgaum based SenseGiz Inc. has come up with STAR, an affordable multi-purpose smartdevice that can be clipped to clothing or worn as a band. This tiny device can track the user's fitness and sleep quality, provide idle alerts, display time, deliver calls, texts or mobile app notifications, control smartphone functions with simple gestures, trigger automatic alarm on loved ones' phone numbers if they suffer a fall or crash, and has a panic button to get help in an emergency.

## Innovative functionalities

STAR was developed with the idea of getting timely help in case of emergencies. Instead of the user taking out the phone and making a call for help, this device automatically triggers an alarm to the user-defined phone numbers. Also, the user does not have to take out the phone every time a call, text or app notification comes up; it can be viewed on the screen of this device.

STAR can also function as a pedometer

and a sleep tracker. It motivates the user to be more active in the day with idle alerts and fitness goal reminders. It tracks physical activities in terms of number of steps taken, distance covered and calories burnt. It also measures sleep quality in terms of light sleep versus deep sleep and interruptions.

Another interesting feature of this device is that, you can define hand gestures (for example, claps, taps or rotation of hands) to perform different operations on your smartphone such as making a call, clicking a photo, playing a song or launching an app.

## What powers the device

The entire circuitry of STAR is enclosed in a polycarbonate casing with a small factor of 40mm (l) x 22mm (w) x 11.5mm (h). It is splash-resistant, though not waterproof, which means it can withstand sweat, splashes and rain. An organic light emitting diode (OLED) display unit serves as the user interface (UI), and the device operates at a temperature range of -20°C to +60°C.

**Power source.** Power to the device is provided by a rechargeable 3.7V, 150mAh lithium-ion polymer (LiP/LiPo/li-poly) battery. When fully charged, the battery lasts for a week even with constant usage. It takes just 90 minutes to charge the device from zero to 100 per cent with the standard micro-USB B charger cable that comes with the product.

**Communication.** STAR uses Bluetooth Smart, also known as Bluetooth 4.0 and Bluetooth Low Energy (BLE), module to connect and sync with the phone app.

**Sensors and processor.** The device houses multiple accelerometers capable of detecting up to 200g force span and a 3-axis gyroscope. The heart of the device is a high-performance microcontroller (MCU) that processes data from sensors to obtain required results.

SenseGiz STAR

